USAMA **ALI**

AI Engineer · ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience in language model training and building AI applications using Python, RAG pipelines, LLM APIs, embeddings, and semantic search. Experienced in FastAPI backend development, database integration, and developing practical AI solutions involving retrieval, research, and intelligent agents.

TOOLS & TECHNOLOGY

- **Programming Languages:**
  Python, C++, SQL
- **AI & Machine Learning:**
  LLMs, GLM Training, RAG, AI Agents, LangChain, LangGraph, Fine-Tuning, Transformers, NLP, Neural Networks, Supervised and Unsupervised Learning, Hyperparameter Tuning, Model Evaluation, EDA, Speech-to-Text, Multimodal AI
- **Backend & Security:**
  FastAPI, REST APIs, SQLAlchemy, Alembic, JWT Authentication, Authorization
- **Databases & Vector Search:**
  PostgreSQL, SQLite, MySQL, pgvector, ChromaDB, FAISS
- **Tools & Infrastructure:**
  Git, GitHub, VS Code, Jupyter Notebook, Ollama, n8n, MLflow, Docker, CI/CD, AWS EC2 and S3

EXPERIENCE

AI Engineer — Hubble42 *June 2026 – Present*

- Contribute to the training of General Language Models (GLMs) as part of AI model development workflows.
- Work on language model training tasks and gain hands-on experience with LLM development.
- Support AI engineering workflows focused on developing and improving language models.

GenAI Engineer Intern — CypherShield *July – November 2025*

- Contributed to GenAI application workflows by integrating and working with LLM APIs.
- Built automation workflows using n8n to streamline recurring development tasks.
- Collaborated with team members to communicate requirements and support GenAI development activities.

PROJECTS

Mindly — AI Knowledge Management (Final Year Project)

- Developing an AI-powered knowledge workspace that transforms content from documents, videos, meetings, and notes into searchable, structured knowledge.
- Working with Python, FastAPI, LLM APIs, embeddings, and retrieval workflows to support semantic search and context-aware answers.
- Website: mindly.one

RepoGuide — AI-Powered Codebase Q&A

- Built repository ingestion, code-aware chunking, embeddings, semantic search, and reranking using Python, FastAPI, PostgreSQL/pgvector, Gemini, Jina AI, and LangChain.
- Implemented database persistence with SQLAlchemy and Alembic migrations for users, repositories, conversations, and messages.
- GitHub: RepoGuide

TutorDesk — AI Teaching Assistant

- Built a RAG-based teaching assistant using Python, FastAPI, Gemini, and structured LLM outputs for course questions, quizzes, assignments, lesson plans, and explanations.
- Implemented document ingestion, PDF extraction, chunking, Gemini embeddings, and course-specific retrieval using PostgreSQL with pgvector.
- Developed a research agent that routes requests between course-material retrieval and web research using LangChain tools and Tavily; integrated a speech-to-text pipeline.
- GitHub: TutorDesk

Movie Recommendation System

- Built a collaborative-filtering recommendation system using Singular Value Decomposition (SVD).
- Implemented data preprocessing, model training, and recommendation generation using Python, Scikit-learn, and Surprise.
- GitHub: Movie Recommendation System

EDUCATION

Bachelor of Science in Computer Science *October 2022 – July 2026*

University of Management and Technology, Lahore